



Course Outline

MATH-1100 CALCULUS (3 Credit Hours)

Pre-Requisite:**Course Title:** Calculus**Course Code:** MATH-1100**Credit Hours:** 3**Instructor:** Mr. Nasir Nadeem**Office:** Mechanical Engineering Department, KFUEIT**Email:** nasir.nadeem@kfueit.edu.pk**Office Hours:** 10:30am - 12:30 pm

Course Objective

This foundational course provides rigorous study in Mathematical Analysis, focusing on theoretical concepts, formal definitions, and proof techniques essential for higher-level quantitative studies, including an introduction to multivariate functions

Course Contents

Foundations of Analysis: Limits, continuity, sequences, and series with emphasis on $\epsilon - \delta$ **proofs** (to satisfy the 'proofs' requirement).

Differential Calculus: Techniques of differentiation, Mean Value Theorem, Taylor/Maclaurin Series.

Vector and Multivariate Calculus: Functions of several variables, Partial Derivatives (First and Second Order), Multiple Integrals (Double and Triple), and basic concepts of Vector Fields.

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes	PLOs	Blooms Taxonomy
1	Students will be able to apply formal mathematical proof techniques to demonstrate the properties of limits and continuity	-	-
2	Students will be able to differentiate and integrate functions of several variables, demonstrating proficiency in Partial Derivatives and Multiple Integration	-	-
3	Students will analyze complex problems using Taylor/Maclaurin Series and apply fundamental theorems of calculus to various domains	-	-

CLO Assessment Mechanism

Assessment tools	CLO_1	CLO_2	CLO_3
Quizzes / Assignments	-	-	-
Mid semester Exam	-	-	-
Final Exam	-	-	-

Overall Grading Policy

Assessment Items	Percentage
Sessional (Quizzes + Assignments + ppts)	20%
Midterm Exam	30%
Final Exam	50%

Text and Reference Books

Text books:

- Calculus: Early Transcendentals by *James Stewart*
- Thomas' Calculus by *George B. Thomas, Jr.* et al

Reference books:

- Introduction to Real Analysis by *Robert G. Bartle*

Administrative Instructions

- According to institute policy, 75% attendance is *mandatory* to appear in the final examination.
- Assignments and report must be submitted as per announced format and within due date.
- For queries, kindly follow the office hours in order to avoid any inconvenience.

Lecture Breakdown

Week 01	Introduction to Real Numbers, Functions, Formal Definition of Limits ($\epsilon - \delta$)
Week 02	Continuity, Inverse Functions, Differential Calculus, Basic Rules of Differentiation
Week 03	Chain Rule, Implicit Differentiation, Related Rates
Week 04	Theorems of Differentiation (Rolle's Theorem, Mean Value Theorem, L'Hôpital's Rule)
Week 05	Integration and Integral Calculus, Definite and Indefinite Integrals, Fundamental Theorem of Calculus.
Week 06	Techniques of Integration (Substitution, By Parts)
Week 07	Applications of Integrals (Area, Volume)
Week 08	Sequences and Infinite Series, Convergence and Divergence Tests, Taylor Expansion
Week 09	Introduction to Multivariate Functions, Level Curves, Limits and Continuity in Higher Dimensions
Week 10	Partial Derivatives (First Order), Chain Rule for Multiple Variables
Week 11	Second Order Partial Derivatives, Mixed Partial Derivatives, Critical Points
Week 12	Multiple Integrals (Double Integrals over Rectangular and General Regions)
Week 13	Change of Variables in Multiple Integrals (Polar, Cylindrical coordinates)
Week 14	Introduction to Vector Fields, Applications of Calculus to Mathematical and Computational Problems



Course Outline

MATH-2122 Differential Equations (3 Credit Hours)

Pre-Requisite: Calculus
Course Title: Differential Equations
Course Code: MATH-2122
Credit Hours: 3
Instructor: Ms. Saba Jamil
Office: 2.21DR, 1st Floor, Electrical Engineering Building, KFUEIT, RYK
Email: saba.jamil@kfueit.edu.pk
Office Hours: 13:30 - 14:30

Course Objective

This course extends the concepts of Calculus by introducing advanced analytical techniques to model and solve real-world problems. It covers higher-dimensional systems and theoretical foundations essential for understanding complex mathematical models

Course Contents

First Order ODEs: Analytical solutions (Separable, Exact, Linear).
Higher Order ODEs: Homogeneous and non-homogeneous equations.
Vector and Higher-Dimensional Systems: Systems of Linear Differential Equations, Matrix methods for solving systems (proofs of existence and uniqueness of solutions are a key focus).
Partial Differential Equations (PDEs): Introduction to first-order PDEs, boundary value problems, and Fourier Series.

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes	PLOs	Blooms Taxonomy
1	Students will be able to apply advanced analytical methods to solve ordinary and systems of differential equations	-	-
2	Students will develop the ability to theoretically analyze the behavior of solutions, including stability and uniqueness (demonstrating analysis with proofs)	-	-
3	Students will be able to formulate simple real-world models using differential equations, including an introduction to higher-dimensional modeling	-	-

CLO Assessment Mechanism

Assessment tools	CLO_1	CLO_2	CLO_3
Quizzes / Assignments	-	-	-
Mid semester Exam	-	-	-
Final Exam	-	-	-

Overall Grading Policy

Assessment Items	Percentage
Sessional (Quizzes + Assignments + ppts)	20%
Midterm Exam	30%
Final Exam	50%

Text and Reference Books

- **Text book:** Elementary Differential Equations and Boundary Value Problems by *Boyce and DiPrima*
- **Reference book:** Differential Equations by *Zill*

Administrative Instructions

- According to institute policy, 75% attendance is *mandatory* to appear in the final examination.
- Assignments and report must be submitted as per announced format and within due date.
- For queries, kindly follow the office hours in order to avoid any inconvenience.

Lecture Breakdown

Week 01	First-order ordinary differential equations (ODEs), Classification, Separable equations, solution methods, and First-Order Linear Equations (Integrating Factors)
Week 02	Solutions to First Order Linear Equations (Integrating Factors)
Week 03	Exact Equations and Numerical Methods for First Order ODEs
Week 04	Second-order linear differential equations, Homogeneous and non-homogeneous equations
Week 05	Method of Undetermined Coefficients and Variation of Parameters
Week 06	Series Solutions for ODEs (Taylor and Power Series)
Week 07	Laplace transforms and their application to solving differential equations
Week 08	Systems of Linear Differential Equations, Matrix Formulation
Week 09	Eigenvalues and Eigenvectors for systems of ODEs
Week 10	Phase Plane Analysis and Stability of systems
Week 11	Introduction to Partial Differential Equations (PDEs), Classification
Week 12	Solutions of simple PDEs (Separation of Variables)
Week 13	Fourier Series and Boundary Value Problems
Week 14	Applications of Differential Equations in Complex Modeling and Data Analysis
Week 15	Numerical Methods for Solving Differential Equations and Error Analysis
Week 16	Presentations



Tentative Weekly Course Plan

Course Title	Linear Algebra		
Course Code	MATH-2103	Credit Hour	3
Name	Mr. Ubaid-ur-Rehman Asghar	Email	ubaid.rehman@kfueit.edu.pk

Course Contents (Core Topics):

- Systems of Linear Equations and Matrices: Gaussian Elimination, Matrix Inverses, Determinants (with proof of properties).
- Vector Spaces and Subspaces: Formal definition of Vector Spaces, Subspaces, Linear Span, Linear Independence and Dependence (with proofs of theorems).
- Basis and Dimension: Formal proof of Existence of Basis, Change of Basis, Rank and Nullity Theorem.
- Linear Transformations: Formal definition and properties of Linear Transformations, Kernel and Range (with proofs).
- Eigenvalues and Eigenvectors: Calculation and theoretical properties of Eigenvalues and Eigenvectors, Diagonalization.
- Inner Product Spaces: Orthogonalization (Gram-Schmidt process) and Projection.

Resource Material	1. Reference Books	
	i	Linear Algebra and Its Applications by <i>David C. Lay</i> or <i>Gilbert Strang</i>
	ii	Elementary Linear Algebra (Applications Version) by Howard Anton
	2. Web Resources	
	i.	
	ii.	

TOPICS TO BE COVERED

Week #	Topics (outline of main topics and sub topics)	Tutorial /Laboratory
1	Real and complex numbers, Introduction to Matrices and Systems of Equations	Tutorial
2	Gaussian Elimination, Gauss-Jordan Elimination, Matrix Operations	Tutorial
3	Determinants and their Theoretical Properties (Proof of basic theorems)	Tutorial
4	Formal Definition of Vector Spaces and their Subspaces	Tutorial



5	Linear Combination, Span, and Proof of Linear Independence/Dependence	Tutorial
6	Basis and Dimension of a Vector Space	Tutorial
7	Row Space, Column Space, and Null Space	Tutorial
8	Linear Transformations and Linear Mappings	Tutorial

9	Definition, Matrix Representation, Properties (Kernel and Range)	
Mid Term Exam		
10	Inner Product Spaces, Orthogonality, Orthogonal Complements	Tutorial
11	Gram-Schmidt Orthogonalization Process	Tutorial
12	Eigenvalues and Eigenvectors (Calculation and Geometric Interpretation)	Tutorial
13	Diagonalization and its applications	Tutorial
14	Mathematical reasoning and Applications of Linear Algebra relevant to Computer Science	Tutorial
15	Review and Introduction to Symmetric Matrices	Tutorial
16	Quadratic Forms (connecting to optimization)	
Final Term Exam		

Student Evaluation criteria:

Assignments	5%
Surprise Test, Quizzes	5%
Presentation	10%
Mid Term Paper	30%
Final Term paper (Marks Distribution: 20 % from Mid Term and 80% from Final Term), i.e. 10 + 40=50	50%
Total	100%

Student Responsibilities:

A student must have attended at least **75%** of the classes held in a course in order to be allowed to sit in the final examination. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beeper must be Turned off (Proper classroom decorum [behavior] adopts, Course outlines and calendars explain requirements and assignments. Students are also responsible for doing all assigned work on time. Students may prepare Sketchbook for taking notes and for references.



Programming Fundamentals + Lab					
Credit Hours:	3	Course Code	COSC-1101 COSC-1201	Instructor	Ms. Sameen Fatima
Objectives:					
This course provides a comprehensive introduction to the core concepts of high-level programming, using the C++ language to focus on algorithmic thinking, structured programming, and the fundamental syntax required for advanced application development					
Course Learning Outcomes:					
- Students will be able to design, implement, and debug basic programs in C++ using structured programming principles. - Students will demonstrate proficiency in fundamental programming constructs including data types, control flow structures, and functions. - Students will be able to analyze simple problems and develop efficient algorithms using arrays and strings to manage data.					
Course Contents:					
- C++ Language Basics: Syntax, Compilation process, fundamental I/O operations. - Data Types and Operators: Primitive and derived data types, precedence, and type casting. - Control Flow: Selection (if-else, switch) and Repetition (for, while) structures. - Functions, Modularity & Recursion: User-defined functions, Parameter passing (by value, by reference), Scope rules, Function overloading, Modularization, and Recursion - Functions and Modularity: User-defined functions, parameter passing (by value, by reference), and scope. - Arrays (single and multi-dimensional), Strings, and basic searching/sorting algorithms. - Analysis & Problem Solving: Basic problem solving, algorithm implementation, and Complexity & Runtime Analysis. - Pointers and Memory: Introduction to pointers and dynamic memory allocation.					
Text Books:					
1. C++ How to Program by <i>Paul Deitel & Harvey Deitel</i> .					
Recommended Books:					
1. C++ Primer by <i>Stanley B. Lippman et al</i>					

Weekly Plan of “Programming Fundamentals” by Ms. Sameen Fatima

Sr. No	Weekly	Topics
1	Week 1st	Introduction to Programming Paradigms, C++ Environment Setup, Compilation vs. Interpretation
2	Week 2nd	C++ Syntax, Variables, Constants, Basic Data Types (int, float, char)
3	Week 3rd	Arithmetic, Relational, and Logical Operators, Type Casting
4	Week 4th	Selection Structures I: if, if-else, and Nested if-else statements
5	Week 5th	Selection Structures II: switch statement and Conditional Operator
6	Week 6th	Repetition Structures I: while loop, do-while loop
7	Week 7th	Repetition Structures II: for loop, Nested Loops, Loop Control (break, continue)
8	Week 8th	Defining and calling functions, Prototypes, Libraries
9	Week 9th	Functions II: Parameter Passing (Pass by Value vs. Pass by Reference), Function Overloading and Recursion
10	Week 10th	Scope Rules (Local vs. Global), Storage Classes (Auto, Static, Extern)
11	Week 11th	Arrays I: Single-Dimensional Arrays, Reading/Writing array data
12	Week 12th	Arrays II: Multi-Dimensional Arrays, Passing Arrays to Functions, Complexity and runtime analysis
13	Week 13th	Strings: C-style Strings and String Manipulation Functions
14	Week 14th	Pointers: Introduction, Pointer Arithmetic, Dynamic Memory Allocation (Basic concepts)
15	Week 15th	File Handling (Input/Output with files) and Debugging and Testing.
16	Week 16th	Introduction to Classes and Objects (OOP Concepts as a foundation for next course)



Object Oriented Programming					
Credit Hours:	3	Course Code	COSC-1102	Instructor	Mr. Muiz-u-din
Objectives:					
This course introduces the fundamental concepts of object-oriented programming using Java. It focuses on designing software using classes and objects and applying core OOP principles such as encapsulation, inheritance, polymorphism, and abstraction.					
Course Learning Outcomes:					
- Students will be able to design and implement programs using classes and objects in Java. - Students will demonstrate understanding of core OOP concepts including encapsulation, inheritance, polymorphism, and abstraction. - Students will be able to use constructors, methods, and access modifiers effectively. - Students will develop modular, reusable, and efficient programs using Java. - Students will apply exception handling and file handling in Java applications.					
Course Contents:					
- Introduction to OOP and Java basics: JVM, syntax, modular program design, and program structure - Classes and Objects: Defining classes, creating objects, access modifiers - Methods and Constructors: Method overloading, constructors - Encapsulation and Abstraction: Data hiding and interfaces - Inheritance: Types of inheritance, code reusability, and reusable components - Polymorphism: Method overloading and method overriding - Interfaces and Abstract Classes - Exception Handling: try-catch, finally, throw/throws - File Handling: Reading and writing files in Java - Arrays and Strings in Java					
Text Books:					
Java: How to Program by <i>Deitel & Deitel</i> - Head First Java by <i>Kathy Sierra & Bert Bates</i>					
Recommended Books:					
Effective Java by Joshua Bloch					

Weekly Plan

Sr. No	Weekly	Topics
1	Week 1st	Introduction to Java and OOP concepts
2	Week 2nd	Java syntax, variables, and data types
3	Week 3rd	Classes and objects
4	Week 4th	Methods and constructors
5	Week 5th	Encapsulation and access modifiers
6	Week 6th	Inheritance basics
7	Week 7th	Types of inheritance and implementation
8	Week 8th	Polymorphism (overloading & overriding)
9	Week 9th	Abstract classes and interfaces
10	Week 10th	Exception handling
11	Week 11th	Arrays and strings
12	Week 12th	File handling
13	Week 13th	Packages, Java libraries, modular program design, and reusable components
14	Week 14th	GUI basics
15	Week 15th	Project development
16	Week 16th	Final revision and evaluation



Course description

COSC-4302 Final Year Project - II (3 Credit Hours)

Pre-Requisite: Final Year Project - I

Course Title: Final Year Project - II

Course Code: COSC-4302

Credit Hours: 3

Instructor: Mr. Mutiullah

Office: 3.2L, 3rd Floor, Institute of Computer Science, ICT Building, KFUEIT, RYK

Email: mutiullah.jamil@kfueit.edu.pk

Office Hours: 10:30 - 11:30

Course Objective

This course is the practical continuation of Final Year Project - I and is designed to help students implement, test, and deploy a complete working solution based on the approved project proposal. Students will develop all core functionalities of their system, integrate different modules, refine the user interface, solve real-world implementation issues, and prepare a final product that demonstrates technical, analytical, and problem-solving skills. The course emphasizes practical development, documentation, teamwork, and final project presentation.

Course Contents

- Development of the proposed system using appropriate tools, technologies, frameworks, and methodologies.
- Connecting all core components and ensuring smooth system workflow.
- Implementation of all required features, operations, and user interactions.
- Unit testing, integration testing, error fixing, and performance improvement.
- Designing and refining the front-end for usability and responsiveness.
- Project presentation, Viva, and prototype demonstration.

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes	PLOs	Blooms Taxonomy
1	Students will be able to implement the approved final year project into a functional working prototype.	-	-
2	Students will be able to integrate all major modules and complete required features of the system.	-	-
3	Students will be able to test, debug, and improve the project for better performance and reliability.	-	-
4	Students will be able to present, document, and defend their final project professionally.	-	-

CLO Assessment Mechanism				
Assessment Tools	CLO_1	CLO_2	CLO_3	CLO_4
Supervisor Evaluation (Meetings, Progress, Implementation)	✓	✓	✓	-
Final Presentation & Demonstration	✓	✓	✓	✓
Viva / Question-Answer Session	-	-	✓	✓

Overall Grading Policy	
Assessment Items	Percentage
Supervisor Evaluation (Progress, Meetings, Implementation)	30%
Final Presentation & Viva (Evaluation by Panel)	70%

Administrative Instructions
- Students must maintain regular contact with their supervisor and submit progress reports on time. - The project must be implemented practically with working functionality and proper source code. - All deadlines for prototype submission, documentation, and final demo must be strictly followed. - Attendance and participation in project meetings are mandatory as per institute policy. - Students must be prepared for final viva, demonstration, and question-answer session.

Lecture Breakdown	
Week 01	Project plan review, scope finalization, and development roadmap
Week 02	Requirement analysis and module-wise implementation planning
Week 03	Database design, project architecture, and environment setup
Week 04	Core functionality development – Part 1
Week 05	Core functionality development – Part 2
Week 06	User interface implementation and refinement
Week 07	Module integration and initial testing
Week 08	Debugging, error handling, and system improvement
Week 09	Additional feature development and optimization
Week 10	Security, validation, and performance enhancement
Week 11	Progress review and prototype demonstration
Week 12	Report writing and technical documentation
Week 13	User manual and final documentation preparation
Week 14	Final code cleanup and system testing
Week 15	Final project rehearsal and presentation preparation
Week 16	Final demo, Viva, and submission



Course title: Object Oriented Programming-Lab					
Credit Hours	1	Course Code	COSC-1202	Instructor	Mr. Muizzud din
Duration	16 weeks	Focus Language	Java	Practical Hours	48
Objectives:					
This module focused on developing practical OOP proficiency by implementing advanced programming concepts (Encapsulation, Inheritance, Polymorphism) in Java, crucial for building modular and scalable software systems.					
Sr. No	Weekly	Topics			
1	Week 01-02	Java Environment Setup (JDK, IDE). Writing and compiling basic Java programs, understanding the main method			
2	Week 03-04	Classes and Objects ki creation: Practical design aur implementation of classes, constructors, aur methods			
3	Week 05	Encapsulation: Data protection using access modifiers (private, public) aur getter/setter methods.			
4	Week 06	Arrays of Objects and Object initialization: Complex data structures handling			
5	Week 07-08	Creating sub-classes aur super-classes, understanding method overriding.			
6	Week 09	Polymorphism: Run-time and compile-time polymorphism (Method Overloading).			
7	Week 10	Abstract Classes and Interfaces implementation: Designing templates for class behavior.			
8	Week 11-12	Exception Handling (Try-Catch-Finally)			
9	Week 13	File Input/Output (I/O) using Java streams.			
10	Week 14	Collections Framework basics: ArrayLists and HashMaps			
11	Week 15-16	Comprehensive OOP Project: Practical software implementation of a small system (Student record system) using Inheritance, Polymorphism, and Encapsulation			



Course title: Data Structures & Algorithms + Lab					
Credit Hours	1	Course Code	COSC-2101 COSC-2201	Instructor	Mr. Saqib Ubaid
Duration	16 weeks	Focus Language	Java	Practical Hours	48
Objectives:					
This module focused on algorithmic problem solving, practical implementation and empirical analysis of complex data structures and fundamental algorithms in Java. Emphasis was placed on solving advanced computational problems, particularly in graph theory (Shortest Path, MST), and comparing algorithm efficiency.					
Sr. No	Weekly	Topics			
1	Week 01-02	Practical implementation of Abstract Data Types (ADTs) using Arrays/Vectors. Introduction to empirical Time Complexity analysis, determinism and termination			
2	Week 03	Implementation of Linked Lists (Singly, Doubly): Creating Nodes and managing dynamic memory allocation			
3	Week 04	Implementation of Stacks and Queues using both arrays and linked lists. Practical applications like Expression Evaluation			
4	Week 05	Sorting Algorithms (1 of 6): Practical implementation of Bubble Sort and Selection Sort and comparative analysis			
5	Week 06	Sorting Algorithms (1 of 6): Practical implementation of Bubble Sort and Selection Sort and comparative analysis			
6	Week 07	Sorting Algorithms (3 of 6): Implementation of Merge Sort and Quick Sort with emphasis on recursion and efficiency			
7	Week 08	Trees basics: Creating Binary Tree Nodes and implementing fundamental Traversal Algorithms (Inorder, Preorder, Postorder)			
8	Week 09	Implementation of Binary Search Trees (BSTs) and AVL Trees: Practical exercises on Insertion, Deletion, and Searching operations			
9	Week 10	Graph Structures implementation: Using Adjacency Matrix and Adjacency List to represent graph data			
10	Week 11	Graph Traversal Algorithms: Practical implementation of Breadth-First Search (BFS) and Depth-First Search (DFS)			
11	Week 12	Shortest Path Algorithm (1): Implementation of Dijkstra's Algorithm (including the use of min-heap/priority queue)			
12	Week 13	Minimum Spanning Tree (MST) Algorithm (1): Implementation of Kruskal's Algorithm (including Union-Find structure)			
13	Week 14	Minimum Spanning Tree (MST) Algorithm (2): Implementation of Prim's Algorithm (using Priority Queue)			
14	Week 15	Implementation of Hashing and Hash Tables including collision resolution techniques			
15	Week 16	Final Lab Project: Algorithmic problem solving & development of a complex problem solution			



Tentative Weekly Course Plan

Course Title	Design & Analysis of Algorithms	Instructor	Mr. Mutiullah Jamil
Course Code	COSC-3115	Credit Hour	3

Objective:

The primary objective of this course is to equip students with the necessary theoretical foundation and practical skills required to design, implement, and rigorously analyze modern algorithms. Students learn to assess the efficiency (time and space complexity), correctness, and computational limits of algorithms, enabling them to devise optimal solutions for complex computational problems.

Course Learning Outcomes:

CLO 1: Analysis: Mathematically analyze the time and space complexity of algorithms using asymptotic notations and solve recurrence relations.

CLO 2: Design: Apply major algorithmic design paradigms, including Divide and Conquer, Greedy Approaches, and Dynamic Programming, to solve various computational problems.

CLO 3: Implementation: Implement standard and advanced algorithms efficiently in a programming language, ensuring both performance and correctness.

CLO 4: Complexity: Classify computational problems based on their inherent difficulty, with a basic understanding of P vs NP and the concept of NP-Completeness.

Reference Books

- *Introduction to Algorithms* by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein (CLRS).
- *Algorithm Design* by Jon Kleinberg and Éva Tardos.

Sr. No	Weekly	Topics
1	Week 01-02	Algorithm definition, Role of algorithms in computing, Algorithm design and analysis, Growth of Functions, Asymptotic Notations and complexity, correctness and efficiency analysis
2	Week 03	Methods for solving recurrence relations: Substitution Method, Recursion Tree, and the Master Theorem
3	Week 04-05	Analysis and implementation of Insertion Sort, Merge Sort (Divide and Conquer), and Heapsort (using Priority Queues)
4	Week 06	Quicksort, Randomized Algorithms, and establishing the lower bounds for sorting

5	Week 07-08	Introduction to Greedy Strategy, Proof of Correctness, Activities Selection Problem, Huffman Coding, Minimum Spanning Trees (Prim's & Kruskal's)
6	Week 09-10	Principles of Dynamic Programming, optimization techniques, Optimal Substructure, Overlapping Subproblems, Examples: Matrix Chain Multiplication, Longest Common Subsequence (LCS), Knapsack Problem
7	Week 11	Graph Representations, Graph Traversal Techniques: Breadth-First Search (BFS) and Depth-First Search (DFS), Topological Sort
8	Week 12	Dijkstra's Algorithm, Bellman-Ford Algorithm, and its application in negative weight cycles
9	Week 13	The Floyd-Warshall Algorithm
10	Week 14	Advanced data structures for efficiency: B-Trees (overview), Hash Tables, and Disjoint Sets
11	Week 15	Decision Problems, Complexity Classes P and NP, NP-Completeness and Reductions
12	Week 16	Introduction to Approximation Algorithms (e.g., Vertex Cover), local search, branch-and-bound, or specific algorithms relevant to Data Science (e.g., Fast Fourier Transform)

Student Evaluation criteria:

Student Evaluation criteria:	5%
Surprise Test, Quizzes:	5%
Semester Project:	10%
Mid Term Paper:	30%
Final Term paper (Marks Distribution: 20 % from Mid Term and 80% from Final Term), i.e. 10 + 40=50:	50%
Total:	100%

Student Responsibilities:

A student must have attended at least 75% of the classes held in a course in order to be allowed to sit in the final examination. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beeper must be Turned off (Proper classroom decorum [behavior] adopts, Course outlines and calendars explain requirements and assignments. Students are also responsible for doing all assigned work on time. Students may prepare Sketchbook for taking notes and for references.



Course Outline

MATH-2124 Numerical Analysis (3 Credit Hours)

Course Title: Numerical Analysis
Course Code: MATH-2124
Credit Hours: 3
Instructor: Ms. Huba Rana
Office: Mechanical Engineering Department, KFUEIT
Email: huba.rana@kfueit.edu.pk
Office Hours: 14:30am - 15:30 pm (Tuesday)

Course Objective

This course is to introduce students to the mathematical concepts and computational techniques required to find approximate solutions to complex mathematical problems that lack analytical (exact) solutions. The course emphasizes understanding the algorithms, error analysis, and stability of numerical methods

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes	PLOs	Blooms Taxonomy
1	Root Finding: Apply and analyze various numerical methods (e.g., Bisection, Newton-Raphson) to efficiently find the roots of non-linear equations.	-	-
2	Solve systems of linear equations using direct (e.g., Gaussian Elimination) and iterative (e.g., Jacobi, Gauss-Seidel) techniques and understand their computational efficiency.	-	-
3	Approximation & Interpolation: Use techniques like Lagrange and Newton interpolation to approximate complex functions, and apply the concept of Least Squares Approximation to fit data (fundamental in Machine Learning).	-	-
4	Calculus Numeric: Implement numerical methods for integration (Quadrature) and differentiation, and apply methods for solving Ordinary Differential Equations (ODEs) like the Runge-Kutta method		

CLO Assessment Mechanism

Assessment tools	CLO_1	CLO_2	CLO_3
Quizzes / Assignments	-	-	-
Mid semester Exam	-	-	-
Final Exam	-	-	-

Overall Grading Policy

Assessment Items	Percentage
Sessional (Quizzes + Assignments + ppts)	20%
Midterm Exam	30%
Final Exam	50%

Text and Reference Books

Recommended books:

- Numerical Analysis by *Richard L. Burden and J. Douglas Faires*
- Numerical Methods for Engineers by *Steven C. Chapra and Raymond P. Canale*
- Introduction to Numerical Analysis by *S. S. Sastry*

Administrative Instructions

- According to institute policy, 75% attendance is *mandatory* to appear in the final examination.
- Assignments and report must be submitted as per announced format and within due date.
- For queries, kindly follow the office hours in order to avoid any inconvenience.

Lecture Breakdown

Week 01-02	Mathematical preliminaries, Sources of Errors (Round-off, Truncation), Absolute and Relative Errors, Stability and Conditioning of problems
Week 03-04	Solutions of non-linear equations (Bisection, Newton-Raphson, Fixed-Point Iteration, Secant Method), Convergence Rates
Week 05-06	Direct & Iterative linear systems (Gauss Elimination, LU Decompositio)
Week 07-08	Direct & Iterative linear systems (Jacobi Method, Gauss-Seidel Method, SOR, Convergence Criteria)
Week 09-10	Lagrange Interpolating Polynomials, Divided Differences, Newton's Form of the Interpolating Polynomial
Week 11	Numerical Differentiation & Integration (Finite Differences, Trapezoidal Rule)
Week 12-13	Numerical Differentiation & Integration (Simpson's Rules, Composite Rules, Gaussian Quadrature)
Week 14	Polynomial Approximation, Linear Least Squares Regression, Non-linear Least Squares
Week 15	Numerical ODEs: Initial Value Problems, Euler's Method
Week 16	Runge-Kutta Methods (RK2, RK4), and Multi-Step Methods



Discrete Structures					
Credit Hours:	3	Course Code	COSC-1103	Instructor	Mr. Mujeeb Ur Rehman
Objectives:					
This course introduces the fundamental abstract and mathematical structures used in Computer Science and Data Modeling. It focuses heavily on formal logic, set theory, graph theory, and proof techniques which are essential for algorithmic analysis and abstract reasoning					
Course Learning Outcomes:					
1. Students will be able to apply formal logic and rigorous proof techniques (including mathematical induction) to solve abstract problems. 2. Students will be able to analyze and model computational problems using the structures of sets, relations, and graphs. 3. Students will develop the abstract reasoning skills necessary for advanced algorithm design and theoretical computer science.					
Course Contents:					
• Foundations of Logic and Proof: Propositional and Predicate Logic, logical equivalences, and formal methods of proof (e.g., Direct Proofs, Proof by Contradiction, Proof by Induction). • Set Theory and Relations: Counting and Combinatorics, Graph Theory, Boolean Algebra, Prime Numbers, Modular Arithmetic, Coding Theory, and Cryptographic Foundations • Counting and Combinatorics: Principles of Counting, Permutations, Combinations, and generating functions. • Graph Theory: Basic definitions of Graphs, Paths, Circuits, Trees, and their properties (with proofs of theorems). • Boolean Algebra and its applications.					
Text Books:					
1. Discrete Mathematics and Its Applications by <i>Kenneth H. Rosen</i>					
Recommended Books:					
1. Discrete Mathematics by <i>Richard Johnsonbaugh</i>					

Weekly Plan of “Discrete Structures” by Mr. Mujeeb Ur Rehman

Sr. No	Weekly	Topics
1	Week 1st	Introduction to Propositional Logic and Compound Propositions,
2	Week 2nd	Conditional Statements, Logical Equivalence
3	Week 3rd	Methods of Proof (Direct Proof, Contraposition, Contradiction)
4	Week 4th	Mathematical Induction
5	Week 5th	Set Theory, Set Operations, Set Identities (with proofs)
6	Week 6th	Functions, Sequences, Summation Notation.
7	Week 7th	Relations, Equivalence Relations, Partial Orderings
8	Week 8th	Number Theory: Prime Numbers, Modular Arithmetic, and Discrete Mathematical Problem Solving
9	Week 9th	Combinatorial Reasoning: Principles of Counting,
10	Week 10th	Permutations, and Combinations
11	Week 11th	Introduction to Graph Theory, Terminology
12	Week 12th	Isomorphism, Trees
13	Week 13th	Eulerian and Hamiltonian Paths, Shortest Path Algorithms
14	Week 14th	Boolean Algebra and Logic Gates
15	Week 15th	Coding Theory and Cryptographic Foundations (Applications of Discrete Mathematics in Security)
16	Week 16th	Presentation



Tentative Weekly Course Plan

Course Title	Database Systems + Lab		
Course Code	COSC-2103 and COSC-2203	Credit Hour	3+1
Name	Mr. Adeel Abid	Email	adeel.abid@kfueit.edu.pk

Objective

This course is to provide students with a comprehensive understanding of the principles, design, and management of Database Management Systems (DBMS). The course focuses on fundamental data models, especially the Relational Model, query languages, and the principles of database integrity and concurrency

Course Learning Outcomes (CLOs)

CLO 1: Data Modeling: Design and develop robust and normalized database schemas using the Entity-Relationship (E-R) Model and map them to the Relational Model.

CLO 2: Querying & Manipulation: Write complex and efficient queries using Structured Query Language (SQL), including joins, sub-queries, and data manipulation commands (DML, DDL).

CLO 3: Normalization: Apply Normalization Theory (up to 3NF and BCNF) to eliminate data redundancy and anomalies, ensuring data integrity and consistency.

CLO 4: Transaction & Control: Explain the concepts of database transactions, concurrency control (e.g., locking), recovery mechanisms, and the basic principles of database security.

1. Reference Books

i *Database System Concepts by Abraham Silberschatz, Henry F. Korth, and S. Sudarshan*

ii *Fundamentals of Database Systems by Ramez Elmasri and Shamkant B. Navathe*

2. Web Resources

i. w3school.com

ii.

Week #	Topics (outline of main topics and sub topics)
1 - 2	Database System Architecture, ANSI-SPARC architecture, Data Independence, DBMS vs. File Systems, Database Users and Administrators
3 - 4	Structure of Relational Databases, Keys (Primary, Foreign), Relational Algebra operations (Selection, Projection, Join, Set Operations), and Relational Calculus
5 - 6	E-R Model Components (Entities, Attributes, Relationships), Mapping Constraints, Enhanced E-R Model (EER)
7 - 8	SQL Data Definition Language (DDL), Data Manipulation Language (DML), Basic Queries, Aggregate Functions, Grouping, and Practical SQL Lab Implementation



9 - 10	Complex Queries, Nested Subqueries, Views, JOIN Operations (Inner, Outer, Cross), and Practical SQL Lab Exercises on Complex Joins & Subqueries
11 - 12	Functional Dependencies, Normalization Forms (1NF, 2NF, 3NF, BCNF), Lossless Join Decomposition
13	Concept of Transactions, ACID Properties (Atomicity, Consistency, Isolation, Durability), Transaction States
14	Need for Concurrency Control, Lock-Based Protocols, Deadlocks, Time-Stamp Ordering Protocol
15	Log-Based Recovery, Shadow Paging, Physical Storage and Data Storage (Files and Records), Indexing (B-Trees)
16	Introduction to NoSQL Databases (Overview), Hierarchical, Network, and Object-Oriented Database Models, Data Warehousing Concepts, Database Security, and Final Database Lab Project

Assignments	5%
Surprise Test, Quizzes	5%
Presentation	10%
Mid Term Paper	30%
Final Term paper (Marks Distribution: 20 % from Mid Term and 80% from Final Term), i.e. 10 + 40=50	50%
Total	100%

Student Responsibilities:

A student must have attended at least **75%** of the classes held in a course in order to be allowed to sit in the final examination. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beeper must be Turned off (Proper classroom decorum [behavior] adopts, Course outlines and calendars explain requirements and assignments. Students are also responsible for doing all assigned work on time. Students may prepare Sketchbook for taking notes and for references.



Course Title: Data Science					
Credit Hours:	3	Course Code	COSC-3109	Instructor	Mr. Maqsood Ahmad
Objectives:					
This course introduce the systematic process of Extracting Knowledge from Data. It emphasizes the foundational steps of the Data Science workflow, including data acquisition, cleaning, statistical analysis, and visualization. The course provides a strong statistical and computational background for handling and interpreting complex datasets, serving as a necessary precursor to advanced Machine Learning studies.					
Course Learning Outcomes:					
• CLO 1: Data Acquisition and Wrangling: Implement practical techniques for data integration, cleaning, and transformation using programmatic tools (e.g., Python/R libraries). • CLO 2: Statistical Foundation: Apply fundamental descriptive and inferential statistics (e.g., probability distributions, hypothesis testing) to explore and summarize data effectively. • CLO 3: Data Visualization: Select and create effective data visualizations (e.g., histograms, scatter plots, heatmaps) to uncover patterns and communicate insights. • CLO 4: Predictive Basics: Understand and implement basic linear modeling concepts and evaluate the quality of simple predictive models.					
Text Books:					
1. Doing Data Science by <i>Cathy O'Neil and Rachel Schutt</i> 2. R for Data Science by <i>Hadley Wickham and Garrett Grolemund</i>					

Weekly Plan of “Data Science” by Mr. Maqsood Ahmad

Sr. No	Weekly	Topics
1	Week 1-2	Data Science pipeline overview, Role of statistics, Data sources and types, data acquisition and preprocessing, Introduction to data manipulation tools with Python (NumPy, Pandas) and R
2	Week 3-4	Data cleaning and missing-value handling (Imputation), Outlier detection, feature engineering, scaling and normalization, Data integration techniques
3	Week 5-6	Exploratory data analysis (EDA), statistical summaries (Mean, Median, Variance), Data summarization, Measures of central tendency and dispersion
4	Week 7-8	Probability Distributions (Normal, Binomial), Central Limit Theorem, Confidence Intervals, Introduction to Hypothesis Testing
5	Week 9-10	Principles of effective visualization using Python (Matplotlib, Seaborn), Univariate and Bivariate visualizations, Plotting time series data, Creating dashboards (basic)
6	Week 11-12	Review of relational data concepts, Basic SQL queries for data extraction and joining tables, Connecting to data sources
7	Week 13-14	Simple Linear Regression, clustering, dimensionality reduction, model evaluation and validation, Evaluating goodness of fit
8	Week 15	Introduction to concepts like Bayes' theorem, Basic decision rules (not algorithms like k-NN/SVM)
9	Week 16	Applying the entire workflow (from raw data to final report) on a real-world dataset



Khawaja Fareed University of Engineering & IT Rahim Yar Khan

Tentative Course Plan **Department of Computer Science**

Class:

Semester:

Session:

Instructor:	Mr. Shahzad Hussain	Email:	shahzad.hussain@kfueit.edu.pk
Course Title:	Machine Learning	Program:	BSCS
Course Number:	COSC-3107	Credit Hours:	3

Lecture Time:	
<u>Course Objective & Outcomes:</u> 1. To introduce students to the basic concepts and techniques of Machine Learning. 2. To develop skills of using recent machine learning software for solving practical problems. 3. To gain experience of doing independent study and research.	
<u>Methods of Teaching:</u> • Class participation is very important and will be evaluated on the bases of class attendance, punctuality, ability to contribute in class discussions, participation in answer questions session and preparation level with assigned readings. All students are expected to prepare for and participate in class. • The methodology of Direct / Live teaching at class room as per Lecture schedule where student have the freedom to interact and ask any question to teacher regarding the current Lecture topic. • Student would also have option to contact with teacher at specific office hours (may change according to summer and winter season) regarding the course related queries.	
Resource Material	1. Recommended & Reference Books
	i Fundamentals of Machine Learning for Predictive Data Analytics, Algorithms, worked examples and case studies 2020, ISBN-10: 0262029448
	ii Practical Machine Learning with Python, 2017; ISBN-10: 1484232062
	iii Introduction to Machine Learning with Python, 2018 ISBN-10: 1449369413
	2. Web Resources
	i. https://machinelearningmastery.com/start-here/
	ii. https://medium.com/analytics-vidhya
Office Help Hours	During Official Hours

SEQUENCE OF TOPICS TO BE COVERED		
Week	Topics (outline of main topics and sub topics)	Book / Tutorial
1	Machine Learning and Applications: Introduction to Machine Learning and Applications: Why you should learn machine learning with us; who this specialization is for and what you will be able to do; getting started with the tools for the course; getting started with Python and Jupyter notebook; getting started with SFrames for data engineering and analysis	
2	Machine Learning Methods Supervised Learning: Classification; Regression. Un-supervised Learning: Clustering, Dimensionality reduction, Anomaly detection, Association rule-mining; Semi-Supervised Learning and Reinforcement Learning. Batch Learning, Online Learning, Instance Based Learning and Model Base Learning.	
3	Machine Learning and Real-world Applications (Use-Case Classification): Analyzing the Sentiment of Reviews; what is intelligent restaurant Review System; Examples of Classification tasks; Linear classifiers; Decision Boundary Evaluating of Classification: Training and evaluating a classifier; model evaluation (Cross-validation, Confusion Matrix, Precision, Recall, ROC-AUC); What a good accuracy; False Positive, false negative; Learning curves; Class prediction Assignment: To implement various Supervised Learning Machine Learning Models on same dataset and for the same task.	
4	Machine Learning and Real-world Applications (Use-Case Regression): Predicting a house price; what good and how might you naively address it? A linear regression: a model-based approach; adding higher order effects. Evaluating regression models: Evaluating overfitting via training / test split; Training / test curves; adding their features; Other regression examples Assignment: To implement various Supervised Machine Learning Regression Models on same dataset and for this task.	
5	The CRISP-DM Process Model Business Understanding, Define Business Problem, Assess and Analysis Scenarios, Define the data-mining problem, Project plan, Data Understanding, Data Collection, Data Description, Exploratory Data Analysis, Data Quality Analysis, Data Preparation, Data Integration, Data Wrangling, Attribute Generation and Selection Implementation CRISP-DM Process Model in real world problem: Real-World Case Study: Predicting Student Grant Recommendations	
6	Processing, Wrangling and Visualizing Data: Data Collection: CSV, JSON, XML, HTML and Scraping, HTML, Web Scraping, SQL, Data Description: Numeric, Text, Categorical Data Wrangling: Understanding Data, Filtering Data, Typecasting, Transformations, Imputing Missing Values, Handling Duplications, Handling Categorical Data, Normalizing Values, String	

	Manipulations.	
7	Processing, Wrangling and Visualizing Data: Data Summarization: Data Visualization: Visualizing with Pandas, Line Charts, Bar Plots, Histograms, Pie Charts, Box Plots, Scatter Plots, Visualization with Matplotlib, Figures and Subplots, Plot formatting, Legends, Axis Controls, Annotations, Global Parameters, Python Visualization Ecosystem.	
8	Feature Engineering and Selection: Features: Understand Your Data Better: Data and Datasets, Features, Models; Feature Engineering on Numeric Data, Feature Engineering on Categorical Data. Feature Engineering on Text Data, Feature Engineering on Temporal Data, Feature Engineering on Image Data	
Mid Exam		
9	Feature Engineering and Selection: Feature Scaling: Standardized Scaling, Min-Max Scaling, Robust Scaling, Feature Selection, Threshold-Based Methods, Statistical Methods, Recursive Feature Elimination, Model-Based Selection, Dimensionality Reduction: Feature Extraction with Principal Component Analysis,	
10	Building, Tuning and Deploying Models: Supervised Learning: Classification and Regression; Generalization, Overfitting and Underfitting; Supervised Machine Learning Algorithms; k-Nearest Neighbors Model: k-Neighbors classification, analyzing KneighboursClassifier, k-neighbors regression, analyzing k-nearest neighbors region, strength, weaknesses and parameters Linear Models: Linear Models: Linear Model for regression and classification; Linear/Logistic Regression; Ridge regression; Lasso; regularization; Linear models for multi-class classification, strength, weaknesses and parameters.	
11	Building, Tuning and Deploying Models: Supervised Learning: Decision Trees: Building Decision Trees; Controlling complexity of Decision Trees & Probabilistic Models: Naive Bayes classification; Building Decision Trees; Controlling complexity of Decision Trees; Analyzing Decision Trees; Feature Importance in trees; Strength, weaknesses and parameters, Ensembles of Decision Trees; Random Forests and Gradient Boosted Regression Trees (GBM). Kernelized Support Vector Machines: Linear Models and Non-linear features, the kernel trick, understanding SVMs, tuning SVM parameters, Preprocessing data for SVMs, strength, weakness and parameters	
12	Building, Tuning and Deploying Models: Supervised Learning: Neural Networks (Deep Learning): the neural network model, tuning neural networks, strength, weaknesses and parameters Classifying movie reviews: a binary classification example	

13	Building, Tuning and Deploying Models: Un-Supervised Learning: Types of unsupervised learning; Preprocessing and Scaling; Dimensionality Reduction, Feature Extraction and Manifold Learning, PCA, NMF, Manifold learning with t-SNE	
14	Building, Tuning and Deploying Models: Un-Supervised Learning: Clustering: k-Means clustering, agglomerative clustering, DBSCAN, comparing and evaluating clustering algorithms; evaluating clustering with ground truth.	
15	Real World Case Studies: Analyzing Bike sharing Trends The Bike sharing Dataset; Problem Statement; Exploratory and Data Analysis; Regression Analysis; Modeling; Testing; Next Steps	
16	Real World Case Studies: Galaxy Classification Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, Deployment	
	Final Term Exam	

Student Evaluation criteria:

Workshop / Assignments/Case study	5%
Surprise Test/Sudden Test , Quizzes	5%
Semester Project	10%
Mid Term Paper	30%
Final Term paper	50%
Total	100%

Student Responsibilities:

Students must attend class. Failure to attend class may result in failure in the course. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beepers must be Turned off (Proper classroom decorum [behavior] adopted. Course outlines and calendars explain requirements and assignments, students are responsible for knowing what they say. Students are also responsible for doing all assigned work on time. Excessive absences (more than 03) will result in “F Grade”. Students may prepare Sketchbook for taking notes and for references.